

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

SSC PART-I (CLASS 9) PRACTICE EXAMINATION

BIOLOGY PAPER 18

Syllabus: Chapters 1 to 7 (revision of the whole course studied so far)

Total Marks: 50

Time Allowed: 2 hours 15 minutes

Version A

HIDAYAAT (Instructions)

1. **Saaray sawal** hal karnay hain. Is paper mein koi bhi **optional sawal nahin** hai.
2. Ye paper sirf **Chapter 1 se Chapter 7** tak ka hai, yani wohi kuch jo aap parh chuki hain. Koi naya chapter is mein shaamil nahin.
3. Jawab apni notebook mein likhein. Har sawal ka number **bilkul waisa hi** likhein jaisa paper par hai, aur tarteeb se hal karein.
4. Section A mein sawal ka number likh kar sirf apnay option ka **letter** likhein, jaise **1. (b)**. Aakhir mein **1 se 10 tak ginti kar lein** ke koi number khali to nahin reh gaya.
5. Jahan **naam** maanga gaya ho wahan angrezi naam likhein. **Naam ya unit na likhne par ½ mark katega**. Har **equation balance** karein.
6. Har explanation aur har jawab ke aakhir mein **ek mukammal jumla** likhein jo sawal ka jawab de.
7. Diagram pencil se bana sakti hain. Baqi paper kaali ya neeli pen se likhein.

SECTION A: MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 MARKS)

*Q.1 Har sawal ka sirf **ek** option sahi hai. Apni copy par number ke saath letter likhein.*

#	Question	A	B	C	D	Answer
1	A scientist is studying the methods of inserting the human insulin gene into bacteria. Which branch of biology may this be?	anatomy	biotechnology	physiology	pharmacy	(A)(B)(C)(D)
2	Colonial organization is the unique feature of the kingdom:	Animalia	Fungi	Protista	Plantae	(A)(B)(C)(D)
3	Which of the following statements correctly represents ribosomes?	made in the nucleus, then work in the cytoplasm, where they make proteins	present only in eukaryotic cells	covered by a single membrane	all of them are attached to the RER	(A)(B)(C)(D)
4	In meiosis the chromosome number of the daughter cells becomes half that of the parent cell. This is caused by the:	replication of DNA during interphase I	crossing over during prophase I	separation of sister chromatids during meiosis II	separation of homologous chromosomes during meiosis I	(A)(B)(C)(D)
5	In Table 5.2 of your book, which organ system keeps the body in balance by removing pathogens, fighting infections and helping in healing ?	lymphatic system	immune system	cardiovascular system	integumentary system	(A)(B)(C)(D)
6	Which structure is at a different level of organization from the other three?	neuron	kidney	liver	lung	(A)(B)(C)(D)
7	Which of the disaccharides is also called the transport sugar?	maltose	fructose	lactose	sucrose	(A)(B)(C)(D)
8	All RNA nucleotides differ from DNA nucleotides in having a different:	nitrogen base	phosphate group	pentose sugar	carboxylic group	(A)(B)(C)(D)
9	In ATP, the two bonds between the phosphate groups, shown by wavy lines, are called:	high energy bonds	hydrogen bonds	peptide bonds	double bonds	(A)(B)(C)(D)
10	What are the products of the light reactions of photosynthesis?	ATP and NADP	ATP, NADPH and oxygen	ATP, PGA and oxygen	PGA and oxygen	(A)(B)(C)(D)

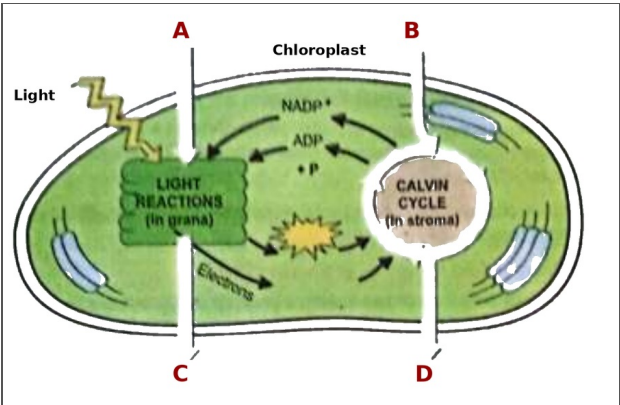
SECTION B: SHORT ANSWER QUESTIONS (6 × 4 = 24 MARKS)

Chhay ke chhay sawal hal karein. Har sawal 4 marks ka hai. Har hissay ke marks bracket mein diye gaye hain.

2.	(Chapter 1) Your textbook uses the biological method to find out how malaria spreads. (i) State the hypothesis that was made about the spread of malaria. (1) (ii) Describe the experiment that Ronald Ross performed on sparrows, and state its result. (2) (iii) How was this hypothesis finally tested on human beings? (1)	4
3.	(Chapter 2) Differentiate between the following, giving two differences for each pair: (i) Bacteria and Protists (2) (ii) Fungi and Plants (2)	4
4.	(Chapter 3) (i) A cell is very rich in smooth endoplasmic reticulum (SER). List any two roles in which this cell will be more efficient. (2) (ii) Which organelles are abundant in a salivary gland cell? Name any two and explain your answer. (2)	4
5.	(Chapter 5) Differentiate between the following, giving two differences for each pair: (i) Xylem and phloem (2) (ii) Dermal tissue and ground tissue of a plant (2)	4
6.	(Chapter 6) (i) Carbohydrates are literally called “hydrates of carbon”. Give the ratio in which carbon, hydrogen and oxygen are present in them, and write their general formula. (2) (ii) In a non-dividing cell the hereditary material is present as chromatin. Write the composition of chromatin, and name the bead like structures that are formed when DNA coils around histones. (2)	4
7.	(Chapter 7) (i) In which part of the cell does glycolysis take place, and what is produced when one molecule of glucose is broken down in it? Write the equation given in your textbook. (2) (ii) Name the four stages of aerobic respiration in the order given in your textbook, and state the stage in which oxygen is used. (2)	4

SECTION C: LONG ANSWER QUESTIONS (2 × 8 = 16 MARKS)

Dono sawal hal karein. Har sawal **8 marks** ka hai.

8.	(Chapter 7) Photosynthesis. The figure below is taken from your own textbook. It shows an overview of photosynthesis inside a chloroplast. Four substances have been replaced by the letters A, B, C and D.  <i>An overview of photosynthesis (from Chapter 7 of your textbook)</i> (i) Name the substances labelled A, B, C and D. (4) (ii) Write the balanced equation of photosynthesis exactly as your textbook prints it, and name the two things written above and below the arrow. (2) (iii) Name the two energy rich molecules that the light reactions make and then hand over to the Calvin cycle. (2)	8
9.	(Chapter 4) Meiosis. (i) Give any four differences between meiosis I and meiosis II. (4) (ii) A cell of an animal contains 18 chromosomes and divides by meiosis. State: (a) the number of chromosomes in each cell at the end of meiosis I, (1) (b) the number of chromosomes in each cell at the end of meiosis II, (1) (c) the total number of daughter cells formed. (1) (iii) Give one reason why the daughter cells produced by meiosis are not genetically identical to one another. (1)	8

END OF QUESTION PAPER

Chapters covered: Ch 1 Science of Biology (MCQ 1, Q.2) | Ch 2 Biodiversity (MCQ 2, Q.3) | Ch 3 The Cell (MCQ 3, Q.4) | Ch 4 Cell Cycle (MCQ 4, Q.9) | Ch 5 Tissues, Organs and Organ Systems (MCQs 5 and 6, Q.5) | Ch 6 Molecular Biology (MCQs 7 and 8, Q.6) | Ch 7 Metabolism (MCQs 9 and 10, Q.7, Q.8).

Student's Name: _____

Date: _____

Marks Obtained: _____ / 50